

Equations with Brackets and Fractions

Distributing over a bracket, clearing a denominator, and equations that need both

Directions

- **Bracket first:** the factor outside multiplies *every* term inside — all of them, not just the first.
 - **Fractions next:** multiply *both whole sides* by the least common denominator. Every term gets multiplied, including the ones with no fraction.
 - Then finish as usual: collect the variable terms on one side, undo the constant, undo the coefficient.
 - Substitute your answer back into the original equation to check it.
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1. Solve: $3(x + 4) = 21$.

Answer: _____

2. Solve: $5(x - 2) = 35$.

Answer: _____

3. Solve: $\frac{x}{4} + 3 = 7$.

Answer: _____

4. Solve: $\frac{2x}{3} = 8.$

Answer: _____

5. Solve: $\frac{x+6}{2} = 5.$

The whole bracket is divided, so the whole of the other side is multiplied.

Answer: _____

6. Solve: $2(3x - 1) + 4 = 26.$

Answer: _____

7. Solve: $\frac{x}{3} + \frac{x}{6} = 3$.

Multiply both sides by the least common denominator of the two fractions first.

Answer: _____

8. Solve: $\frac{x+1}{2} + \frac{x-3}{4} = 5$.

Answer: _____

9. Solve: $4(2x+3) = 3(x+14)$.

Both brackets must be expanded before anything can be collected.

Answer: _____

10. Solve: $\frac{3x}{5} - 2 = \frac{x}{5} + 4.$

Answer: _____

11. Solve: $\frac{2(x-6)}{3} = \frac{x+2}{2}.$

Answer: _____

12. Challenge. Nia solves $3(x - 4) = 18$ like this:

$$3(x - 4) = 18 \longrightarrow 3x - 4 = 18 \longrightarrow 3x = 22 \longrightarrow x = \frac{22}{3}$$

- (a) Solve the equation correctly. Answer: _____
- (b) Explain which step went wrong and what rule Nia broke.